

ASTRA: A Python Package for Cross-Instrument Stellar and Telluric Template Construction

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Summary

ASTRA is a Python package that provides a modular, instrument-independent interface for working with high-resolution stellar spectra. Designed to support data from multiple spectrographs—including ESPRESSO (Pepe et al., 2021), HARPS (Mayor et al., 2003; Pepe et al., 2002), MAROON-X (Seifahrt et al., 2022), and CARMENES (Quirrenbach et al., 2014)—*ASTRA* offers a unified abstraction over their data formats, enabling consistent access to fluxes, wavelengths, uncertainties, and metadata across instruments. Furthermore, it applies the necessary wavelength and flux calibrations that are needed, as described by the official pipelines of each instrument.

In addition to a common interface, *ASTRA* provides internal quality control checks of the observations, automatically divides them into the different sub-datasets that are commonly used by each spectrograph, and provides avenues to dynamically reject observations based on different properties. Furthermore, it also provides routines to mask the spectral imprint of Earth's atmosphere (in the form of telluric lines) and construct high-SNR, data-driven, stellar templates.

This package serves as the backend of the sBART pipeline (Silva et al., 2022) and is designed to be extensible and suitable for integration into larger spectral analysis workflows, enabling the construction of pipelines without having to tailor them to individual instruments. It is implemented in such a way that the user can select to only open in memory a small number of observations, such that it can seamlessly handle datasets with thousands of observations. Furthermore, it makes use of Python's autoproxy objects, ensuring a smooth integration with codes that use the *multiprocessing* library to leverage concurrent processing for faster computations.

Statement of need

In recent years, multiple ultra-stable, high-resolution spectrographs capable of meter-per-second (or better) radial velocity precision have become central to exoplanet and stellar astrophysics, e.g. HARPS, CARMENES, MAROON-X, and ESPRESSO. While each spectrograph provides high-quality observations, they also use distinct data formats and apply different corrections to the stellar spectra. This hinders the development of generalized analysis pipelines, and often leads to analysis pipelines that are focused on a single instrument. To the best of our knowledge, there is no library that provides a similar interface to access stellar spectra of multiple state-of-the-art spectrographs.

41 *ASTRA* was built with the intention of not only providing a common API to access stellar
42 spectra, but also provides:

- 43 1. Management of observations – *ASTRA* provides a high-level interface for accessing stellar
44 spectra and metadata, built on a memory-efficient design that only loads a minimal
45 number of spectra at a time. This ensures that *ASTRA* stays responsive, even when
46 dealing with datasets with thousands of observations, at the cost of computational speed
47 when interfacing with the data. This is done through the *autoprox* interface, ensuring
48 full compatibility with any *multiprocessing*-based application.
- 49 2. Masking of atmospheric features – When dealing with ground-based spectroscopy, it is
50 important to account for the spectral imprint of our atmosphere (in the form of telluric
51 lines), as well as its yearly variation. *ASTRA* can automatically run *TelFit* (Gullikson
52 et al., 2014) to generate a synthetic transmittance model and create a binary mask to
53 reject the position of telluric lines;
- 54 3. Construction of high-SNR stellar models – The construction of high-SNR stellar templates
55 from observations is pivotal for the extraction of precise radial velocities (Artigau et al.,
56 2022; Silva et al., 2022; Zechmeister et al., 2018), determination of stellar parameters
57 (Gomes da Silva, J. et al., 2021; Sousa, S. G. et al., 2024), and characterization of
58 exoplanetary atmospheres (Azevedo Silva et al., 2022; Damasceno, Y. C. et al., 2024;
59 Stangret, M. et al., 2024);
- 60 4. Dynamically group the observations into different sub-datasets – Over their lifetime, most
61 state-of-the-art spectrographs are subjected to instrumental interventions, leading to
62 changes in the instrumental profile and offsets in radial velocities. As a consequence, it is
63 often necessary to divide our data in the time-periods before and after such interventions,
64 to construct individual templates in each. *ASTRA* is pre-configured with the dates of
65 such instruments for all supported spectrographs, divides the observations in each dataset
66 (or sub-instrument) and creates individual stellar and telluric templates for each;
- 67 5. Filtering of observations – When analysing data, we often reject observations based on
68 metadata information (weather conditions, airmass, among others). Within *ASTRA*
69 the user can dynamically set filters on different properties, with the goal of either fully
70 rejecting the observation, or rejecting it for a specific operation. This means that it is
71 possible to reject an observation for the construction of the stellar template, but not
72 reject it from any subsequent analysis.
- 73 6. Masking of wavelength regions – When dealing with stellar spectra we often need to
74 reject wavelength regions due to different contaminating effects. *ASTRA* creates an
75 internal binary mask for all pixels and allows the rejection of i) telluric-contaminated
76 regions; ii) activity-sensitive regions; iii) user-provided wavelength intervals.

77 As the backend of the *sBART* pipeline, it is already in use for scientific production, and is
78 well-positioned to support the broader astrophysical community working with high-resolution
79 spectroscopy.

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References

- Artigau, E., Cadieux, C., Cook, N. J., Doyon, R., Vandal, T., Donati, J.-F., Moutou, C., Delfosse, X., Fouqué, P., Martioli, E., Bouchy, F., Parsons, J., Carmona, A., Dumusque, X., Astudillo-Defru, N., Bonfils, X., & Mignon, L. (2022). Line-by-line velocity measurements: An outlier-resistant method for precision velocimetry. *The Astronomical Journal*, 164(3), 84. <https://doi.org/10.3847/1538-3881/ac7ce6>
- Azevedo Silva, T., Demangeon, O. D. S., Santos, N. C., Allart, R., Borsa, F., Cristo, E., Esparza-Borges, E., Seidel, J. V., Palte, E., Sousa, S. G., Tabernero, H. M., Zapatero Osorio, M. R., Cristiani, S., Pepe, F., Rebolo, R., Adibekyan, V., Alibert, Y., Barros, S. C. C., Bouchy, F., ... Udry, S. (2022). Detection of barium in the atmospheres of the ultra-hot gas giants WASP-76b and WASP-121b: Together with new detections of Co and Sr+ on WASP-121b. *Astronomy & Astrophysics*, 666, L10. <https://doi.org/10.1051/0004-6361/202244489>
- Damasceno, Y. C., Seidel, J. V., Prinoth, B., Psaridi, A., Esparza-Borges, E., Stangret, M., Santos, N. C., Zapatero-Osorio, M. R., Alibert, Y., Allart, R., Azevedo Silva, T., Cointepas, M., Costa Silva, A. R., Cristo, E., Di Marcantonio, P., Ehrenreich, D., González Hernández, J. I., Herrero-Cisneros, E., Lendl, M., ... Pepe, F. (2024). The atmospheric composition of the ultra-hot Jupiter WASP-178 b observed with ESPRESSO*. *Astronomy & Astrophysics*, 689, A54. <https://doi.org/10.1051/0004-6361/202450119>
- Gomes da Silva, J., Santos, N. C., Adibekyan, V., Sousa, S. G., Campante, T. L., Figueira, P., Bossini, D., Delgado-Mena, E., Monteiro, M. J. P. F. G., de Laverny, P., Recio-Blanco, A., & Lovis, C. (2021). Stellar chromospheric activity of 1674 FGK stars from the AMBRE-HARPS sample - i. A catalogue of homogeneous chromospheric activity*. *Astronomy & Astrophysics*, 646, A77. <https://doi.org/10.1051/0004-6361/202039765>
- Gullikson, K., Dodson-Robinson, S., & Kraus, A. (2014). Correcting for telluric absorption: Methods, case studies, and release of the Telfit code. *The Astronomical Journal*, 148(3), 53. <https://doi.org/10.1088/0004-6256/148/3/53>
- Mayor, M., Pepe, F., Queloz, D., Bouchy, F., Rupprecht, G., Lo Curto, G., Avila, G., Benz, W., Bertaux, J.-L., Bonfils, X., Dall, Th., Dekker, H., Delabre, B., Eckert, W., Fleury, M., Gilliotte, A., Gojak, D., Guzman, J. C., Kohler, D., ... Weilenmann, U. (2003). Setting new standards with HARPS. *The Messenger*, 114, 20–24.
- Pepe, F., Cristiani, S., Rebolo, R., Santos, N. C., Dekker, H., Cabral, A., Di Marcantonio, P., Figueira, P., Curto, G. L., Lovis, C., Mayor, M., Mégevand, D., Molaro, P., Riva, M., Osorio, M. R. Z., Amate, M., Manescau, A., Pasquini, L., Zerbi, F. M., ... Zanutta, A. (2021). ESPRESSO@VLT – On-sky performance and first results. *Astronomy & Astrophysics*, 645, A96. <https://doi.org/10.1051/0004-6361/202038306>
- Pepe, F., Mayor, M., Rupprecht, G., Avila, G., Ballester, P., Beckers, J.-L., Benz, W., Bertaux, J.-L., Bouchy, F., Buzzoni, B., Cavadore, C., Deiries, S., Dekker, H., Delabre, B., D'Odorico, S., Eckert, W., Fischer, J., Fleury, M., George, M., ... Penny, A. (2002). HARPS: ESO's coming planet searcher. Chasing exoplanets with the La Silla 3.6-m telescope. *The Messenger*, 110. <https://ui.adsabs.harvard.edu/abs/2002Msngr.110....9P>
- Quirrenbach, A., Amado, P. J., Caballero, J. A., Mundt, R., Reiniers, A., Ribas, I., Seifert, W., Abril, M., Aceituno, J., Alonso-Floriano, F. J., Eiff, M. A., Jiménez, R. A., Anwand-Heerwart, H., Azzaro, M., Bauer, F., Barrado, D., Becerril, S., Béjar, V. J. S., Benítez, D., ... Xu, W. (2014). CARMENES instrument overview. In S. K. Ramsay, I. S. McLean, & H. Takami (Eds.), *Ground-based and airborne instrumentation for astronomy v* (Vol. 9147, p. 91471F). International Society for Optics; Photonics; SPIE. <https://doi.org/10.1117/12.2056453>
- Seifahrt, A., Bean, J. L., Kasper, D., Stürmer, J., Brady, M., Liu, R., Zechmeister, M., Stefánsson, G. K., Montet, B., White, J., Tapia, E., Mocnik, T., Xu, S., & Schwab, C. (2022). MAROON-X: The first two years of EPRVs from Gemini North. In C. J.

- 139 Evans, J. J. Bryant, & K. Motohara (Eds.), *Ground-based and airborne instrumentation*
140 *for astronomy IX* (Vol. 12184, p. 121841G). International Society for Optics; Photonics;
141 SPIE. <https://doi.org/10.1117/12.2629428>
- 142 Silva, A. M., Faria, J. P., Santos, N. C., Sousa, S. G., Viana, P. T. P., Martins, J. H. C.,
143 Figueira, P., Lovis, C., Pepe, F., Cristiani, S., Rebolo, R., Allart, R., Cabral, A., Mehner,
144 A., Sozzetti, A., Mascareño, A. S., Martins, C. J. A. P., Ehrenreich, D., Mégevand, D.,
145 ... al, et. (2022). A novel framework for semi-Bayesian radial velocities through template
146 matching. *Astronomy & Astrophysics*. <https://doi.org/10.1051/0004-6361/202142262>
- 147 Sousa, S. G., Adibekyan, V., Delgado-Mena, E., Santos, N. C., Rojas-Ayala, B., Barros,
148 S. C. C., Demangeon, O. D. S., Hoyer, S., Israelian, G., Mortier, A., Soares, B. M. T.
149 B., & Tsantaki, M. (2024). SWEET-cat: A view on the planetary mass-radius relation.
150 *Astronomy & Astrophysics*, 691, A53. <https://doi.org/10.1051/0004-6361/202451704>
- 151 Stangret, M., Palle, E., Esparza-Borges, E., Orell Miquel, J., Casasayas-Barris, N., Zapatero
152 Osorio, M. R., Cristo, E., Allart, R., Alibert, Y., Borsa, F., Demangeon, O. D. S., Di
153 Marcantonio, P., Ehrenreich, D., Figueira, P., González Hernández, J. I., Herrero-Cisneros,
154 E., Martins, C. J. A. P., Santos, N. C., Seidel, J. V., ... Udry, S. (2024). The obliquity and
155 atmosphere of the hot Jupiter WASP-122b (KELT-14b) with ESPRESSO: An aligned orbit
156 and no sign of atomic or molecular absorption*. *Astronomy & Astrophysics*, 691, A120.
157 <https://doi.org/10.1051/0004-6361/202450938>
- 158 Zechmeister, M., Reiners, A., Amado, P. J., Azzaro, M., Bauer, F. F., Béjar, V. J. S., Caballero,
159 J. A., Guenther, E. W., Hagen, H.-J., Jeffers, S. V., Kaminski, A., Kürster, M., Launhardt,
160 R., Montes, D., Morales, J. C., Quirrenbach, A., Reffert, S., Ribas, I., Seifert, W., ...
161 Wolthoff, V. (2018). Spectrum radial velocity analyser (SERVAL): High-precision radial
162 velocities and two alternative spectral indicators. *Astronomy & Astrophysics*, 609, A12.
163 <https://doi.org/10.1051/0004-6361/201731483>